

Internship Social Activity (DEC-691)

Human Factors–Driven Redesign:

Helm and Integrated Digital Dashboard
for Kochi Water Metro Ferries

*Submitted in the partial fulfillment of the
Requirements for the award of the degree*

Of

Masters in Design

By

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Under the Supervision of

Prof. Unni Mohan M.

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Date: 12th September, 2025

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Ms. Jyoti Rautela**, a student from Indian Institute of Technology Roorkee, has successfully completed the project work titled : **Design of Cockpit for Inland Boats** under the supervision of Prof. Unni Mohan .M, Department of IDC School of Design, during the period from 19th June, 2025 to 19th August, 2025.



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Internship Completion Letter

This is to certify that **Ms. Jyoti Rautela**, a M.Des. student at IIT Roorkee, successfully completed her internship under my guidance at the **IDC School of Design, IIT Bombay**, from **June 19, 2025 to August 19, 2025**.

During this period, she worked on the project titled **"Design of Cockpit for Inland Boats."** She contributed with sincerity and dedication, engaging in design research, ideation, and conceptual development for the project objectives.

I hereby confirm the successful completion of her internship and grant her approval to **showcase and include the work carried out during the internship as part of her academic course requirements.**

We appreciate her contributions and wish her the very best in her future academic and professional pursuits.

Sincerely,

Unni Mohan M

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ABSTRACT

The goal of this project is to remodel the helm and integrated digital dashboard for Kochi Water Metro-type boats using human factors. The existing control panel is not well suited to Indian anthropometry or the requirements of a diverse collection of operators, including female captains. It is also visually cluttered and disperses important controls over several locations. The study uses field observations, ferry operator interviews, secondary research on bridge/helm ergonomics, and analysis of current helm layouts from Kochi and similar ships to remedy the issue.

Primary, secondary, and tertiary reach zones are defined using Human Factors in Design (HFD) principles and Indian anthropometric data. It also determines which functions are moved or reflected on the digital dashboard and which remain as physical buttons. The hierarchy and grouping of digital operations (navigation, engine monitoring, communication, and warnings) are informed by an information architecture mapping and card-sorting exercise.

The suggested solution includes a series of dashboard screen designs, an integrated helm form, and a rearranged physical control layout. All of these components were assessed through ergonomic testing using full-scale mock-ups and peer evaluation. The result is a cockpit-dashboard system for Indian inland water transportation that lowers cognitive burden, increases reach and visibility, and provides a more inclusive and future-ready interface.

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List of Abbreviations

AIS – Automatic Identification System

CAD – Computer-Aided Design

DG Shipping – Directorate General of Shipping

DGPS – Differential Global Positioning System

GPS – Global Positioning System

HFD – Human Factors in Design

HMI – Human–Machine Interface

IA – Information Architecture

IMO – International Maritime Organization

KWM – Kochi Water Metro

MTCC – Maldives Transport and Contracting Company

NM – Nautical Mile

PPE – Personal Protective Equipment

RPM – Revolutions Per Minute

UI – User Interface

UX – User Experience

VHF – Very High Frequency (radio)

DWL – Design Waterline

CL – Centreline (longitudinal centre of the vessel)

FWD – Forward

AFT – Aft (rear part)

STBD – Starboard (right side of vessel, facing forward)

PORT – Port (left side of vessel, facing forward)

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1. Introduction

Inland water transport systems like the Kochi Water Metro rely on a well-designed helm as the primary interface between the captain and the vessel. In the current setup, controls and screens have evolved through incremental additions, leading to a cluttered console, scattered controls, and a digital dashboard that is not fully integrated with the physical layout. This increases cognitive load, slows decision-making, and may not adequately accommodate the diverse body sizes and working conditions of Indian ferry operators.

This project uses Human Factors in Design (HFD) and ergonomics to **redesign both the physical helm and the digital dashboard** as a single, coherent cockpit system. Indian anthropometric data, task analysis, and card-sorting-based information architecture are used to define control placement, reach zones, and digital hierarchy. The aim is to create an integrated, ergonomic, and inclusive helm–dashboard design that improves reach, visibility, comfort, and clarity of information for present and future ferry operators.

1.1 Objective

To design an ergonomically optimised, low-cognitive-load **integrated helm and digital dashboard** for Kochi Water Metro type ferries, focusing on physical placement of steering, throttle, buttons and screens as well as digital information hierarchy for Indian ferry operators (including women captains).

- Reduce physical clutter and ambiguous control placement on the helm.
- Optimise **reach, visibility and comfort** using Indian anthropometric data and HFD principles.
- Design a **digital dashboard** with clear grouping and hierarchy of functions (navigation, engine, communication, alerts).
- Evaluate the integrated design via ergonomic tests, card sorting and peer review.



(a)



(b)

Fig 1.1 (a) Kochi water metro (b) Kochi water metro dashboard & helm (Muziris)

Objective	Indicator	How measured
Reduce clutter on the helm	Fewer control groups / panels	Count groups on existing vs proposed console
Improve reach to key controls	Most frequent controls in primary reach zone	Map controls on plan view and check against reach data
Clarify digital information hierarchy	Fewer steps to access key functions	Count taps/clicks for main tasks on dashboard
Improve ergonomic comfort for operators	Higher average comfort ratings (1–5 scale)	Participant ratings after using full-scale mock-up

Table 1 Objectives vs Measurable Indicators

2. Association with Course Outcomes

This project directly applies Human Factors in Design (HFD), Ergonomics, Interface Design and CAD/Prototyping learnings to a real transportation context (Kochi Water Metro).

2.1 Application of Human Factors and Ergonomics

This project directly applies HFD principles by grounding all helm dimensions in Indian 5th–95th percentile anthropometric data. Steering, throttle, buttons and screens are positioned to minimise stretching and awkward postures in both sitting and standing use. Controls are organised into primary, secondary and tertiary reach zones so that the most frequent and critical actions fall naturally within the operator’s comfortable reach and clear line of sight.

2.2 Application of Interface and Digital Design

Interface design learning is used to structure the digital dashboard through a clear information architecture of navigation, engine, communication and auxiliary functions. Card sorting with peers helps validate how users naturally group and label these functions, informing the final menu and screen structure. Principles of visual hierarchy, consistent iconography and reduced steps for critical tasks guide the layout of key screens and alarm states.

2.3 Application of Studio, CAD and Prototyping

Studio and CAD skills are applied by taking the helm from hand sketches to refined 3D models and physical mock-ups. Form exploration focuses on integrating screens and controls into a single coherent volume that fits the ferry context. Scale models and a full-scale mock-up are then used to evaluate proportions, reach and visibility, enabling iterative refinement based on real-world feedback.

2.4 Integration of Theory and Practice

Across all stages, the project ties classroom theory to a live marine context by combining HFD, interface design and prototyping methods. Each course contributes a specific layer—body fit, information clarity and buildability—to the final helm–dashboard system. This demonstrates how multidisciplinary design learning can converge into a practical, industry-relevant solution.

3. Literature Review and Background

The literature on ship and ferry helm design can be grouped into four main strands:

(i) international standards and guidelines for bridge layout and ergonomic criteria, (ii) human factors and anthropometric data for defining reach and posture, (iii) research on marine human–machine interfaces and digital dashboards, and (iv) case studies and human-centred design approaches to ship bridges. Together, these sources provide the normative requirements, human-scale data and design processes that inform the integrated physical–digital cockpit developed in this project.

3.1 Ferry Helm & Bridge Ergonomics

International standards such as ISO 8468 and IMO MSC/Circ.982 define functional requirements for bridge configuration, workstation arrangement, and ergonomic criteria for bridge equipment and layout, emphasising clear sight lines, functional grouping and operator-centred positioning of controls and displays [1–3].

Guidance documents and reviews on bridge ergonomic design further stress user-centred layouts that support safe navigation, reduce workload, and avoid confusion arising from ad-hoc

equipment additions [4–6]. These works motivate treating the helm as a carefully designed control–display system rather than a collection of independent devices.

3.2 Anthropometry and Reach Zones

Ergonomics and anthropometry literature shows how body dimensions, reach envelopes and eye height should directly inform control placement and working posture [7–9].

Indian anthropometric studies and datasets provide standing and sitting reaches for male and female users and have been used to define gender-neutral reach envelopes for control workplaces, suggesting comfortable horizontal reaches of roughly 0.3–0.4 m for frequent tasks and up to about 0.6–0.7 m for occasional operations [10–12]. These principles underpin the use of primary, secondary and tertiary reach zones for organising steering, throttle, buttons and screens in this project.

3.3 Marine Digital Dashboards and HMI

Recent work on bridge human–machine interaction highlights that adding more technology does not automatically translate to better safety; poorly integrated displays can increase workload, fragmentation and confusion for deck officers [13,14].

Guidelines on bridge equipment and layout, as well as industry bridge layout notes, recommend functionally oriented groupings (e.g., navigation, monitoring, communication) and consistent, ergonomic interfaces that support situation awareness and quick access to critical functions [2,3,15]. These recommendations directly inform the structuring of the digital dashboard into clear functional clusters and shallow navigation hierarchies in the proposed design.

3.4 Case Studies and Human-Centred Ship Design

Recent ferry projects in India and abroad show how **helm layout, digital dashboards, and human-centred bridge design** are evolving together. The following case studies were chosen because they combine **electric / hybrid propulsion, dense urban or fragile natural contexts,** and **explicit emphasis on ergonomics and information hierarchy**. Together they form a comparative backdrop for your own Kochi Water Metro helm and dashboard redesign.

3.4.1 Aditya – India’s First Solar Ferry (Kerala, India)

Aditya is a 20 m × 7 m solar-electric catamaran built by NavAlt for the Kerala State Water Transport Department, carrying about 75 passengers on the Vaikom–Thavanakkadavu route on Vembanad Lake. It is marketed as India’s first commercially viable solar ferry and has won international awards for electric vessel design. [14]

From an HFD perspective, Aditya represents a “minimum intervention” cockpit: a relatively conventional wheelhouse with standard helm, throttle, and navigation instruments adapted to a new propulsion system. The main changes are backend (solar–battery–generator integration) while the operator interface stays familiar, which is useful for comparison with more radical bridge redesigns like the Unified Bridge. [15]



Fig 3.1 Exterior view of Aditya solar ferry

3.4.2 Kochi Water Metro – Electric Urban Water Transit (Kochi, India)

Kochi Water Metro is an integrated electric-hybrid ferry system connecting 10 island communities with 38 terminals and 16 planned routes, fully integrated with Kochi Metro rail. [19] The system uses a fleet of battery-powered boats designed for high frequency (10–20 min headways) and short urban hops, placing strong constraints on turn-around time, visibility at terminals, and quick situation awareness on the bridge. [20]

Current boats show a fairly utilitarian cockpit: helm located on the longitudinal centreline, with multi-function displays, radar, engine monitoring panels, and paper charts often coexisting in a

compact space. This is a good reference case for analysing field of view to terminals, blind zones, and the distribution of physical controls vs. digital screens in a high-traffic urban context. [19]



Fig 3.2 Exterior of Kochi Water Metro vessel

3.4.3 Raajje Transport Link (RTL) – MTCC High-Speed Ferries (Maldives)

The Maldives Transport and Contracting Company (MTCC) operates the **Raajje Transport Link (RTL)**, a high-speed ferry and bus network designed to connect multiple atolls through integrated public transport. The integrated national public ferry network is being rolled out in phases, with high-speed ferries built by Gulf Craft and others to serve as the backbone of inter-island connectivity. [17]

RTL ferries run at higher speeds than typical urban waterbuses and often in open-sea conditions, so **helm layout, forward visibility, and motion ergonomics** become critical. [17]



(a)



(b)

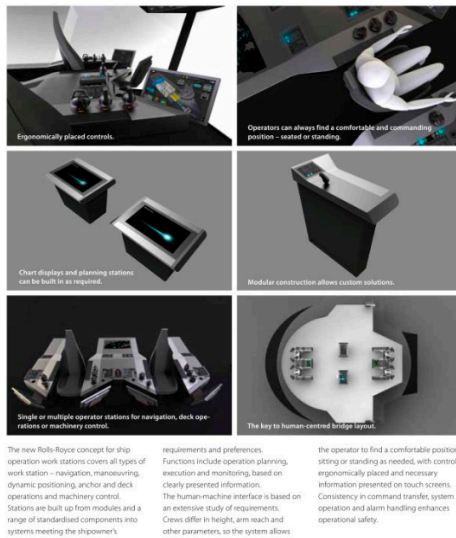
Fig 3.3 (a) Helm / wheelhouse of RTL ferry with skipper at controls, (b) RTL high-speed ferry

3.4.4 Rolls-Royce “Unified Bridge” – Human-Centred Ship Bridge (Offshore Vessels)

The Rolls-Royce Unified Bridge was conceived as a **complete re-thinking of the ship’s bridge** for complex offshore vessels, aiming to simplify an increasingly cluttered environment of disparate panels and displays. The design consolidates controls into an integrated workstation with consistent GUI across systems, large touch displays, and carefully designed seating and visibility to improve safety and reduce operator workload. [18]

This project explicitly applies **human-centred design and ergonomics**: reduced number of control types, harmonised visual language, and a 360° view with optimised sight lines to deck operations. It serves as a strong precedent for your thesis argument that **helm, throttle, screens and alarms must be treated as one holistic “interaction space”**, not as independent pieces bolted on over time. [19]

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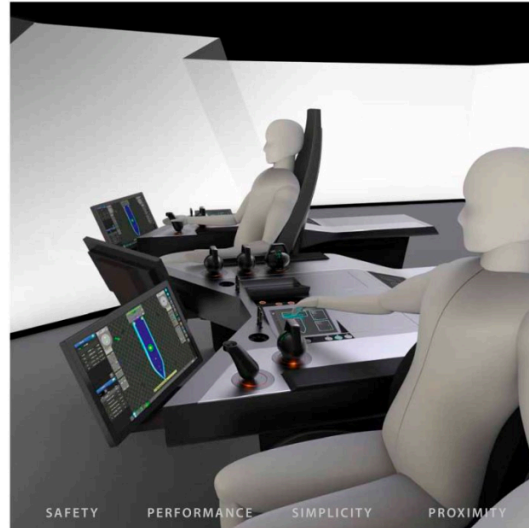


Fig 3.4 Overall Unified Bridge console showing integrated screens and controls

3.4.5 Vision & Future of the Fjords – Inclusive & Sustainable Sightseeing Ferries (Norway)

The **Vision of The Fjords** (diesel-electric hybrid) and **Future of The Fjords** (all-electric) are 42.5 m carbon-fibre passenger catamarans operating in Norway's UNESCO-listed Nærøysfjord. Vision was the world's first hybrid sightseeing ship of its kind, and Future is a fully electric sister ship, designed to minimise noise, emissions and wake. [20]

These vessels are notable for combining **sustainability** with **inclusive passenger experience**: stepped “Seasight” deck geometry gives nearly 360° views, and an inclusive design case study highlights specific provisions for passengers with reduced mobility and vision [20]. The bridge itself follows a **clean, highly glazed wheelhouse with a compact cluster of digital navigation and control systems**, allowing the crew to operate quietly and smoothly in very sensitive natural environments. DNV and other sources emphasise the hull and propulsion as well as operations in low-wake, low-noise mode [20].

3.5 Identified Gaps

Across these strands, authors repeatedly note that many bridges still treat physical controls and digital systems as separate layers, with standards only partially implemented and human factors research not fully translated into concrete layouts [4,6,13]. There is also limited explicit focus on Indian anthropometric profiles and on inclusivity for a wide range of operators, including women captains, in ferry and inland waterway contexts [10–12].

This project addresses these gaps by applying bridge ergonomics standards, Indian anthropometric data and HMI principles together to the specific case of a Kochi Water Metro-type helm and digital dashboard.

4. Outcomes and Deliverables

This project produces a set of interlinked outcomes that translate Human Factors in Design (HFD) principles into a concrete helm–dashboard solution for an Indian inland ferry context. The deliverables span (i) an integrated physical helm architecture, (ii) a digital dashboard information and screen structure, (iii) validated prototypes and evaluation results, and (iv) transferable guidelines for future cockpit development.

5. Context

5.1 Introduction and Project Context

Context: Kochi Water Metro, new ferry systems in India, emerging women pilots.

Problem: Cluttered helm, scattered controls, multiple uncoordinated screens → high cognitive load, risk of errors.

Need: Integrated **physical cockpit + digital dashboard** tailored to Indian anthropometry.

5.2 Project Brief

Design a highly functional, aesthetically pleasing and durable helm and digital dashboard that reduce cognitive load, integrate essential analogue controls with a sophisticated digital console, and adhere to ergonomic and regulatory standards.

5.3 Research Methodology

5.3.1 Secondary Research

Our secondary research combined standards and guidelines on helm/bridge ergonomics with case studies of existing ferries. International and national documents (IMO, DG Shipping, ABS, Nautical Institute, etc.) emphasise a few recurring principles: clear sight lines over the bow, functional grouping of controls, and consistent relationships between controls and displays.

They also stress designing around reach envelopes and anthropometric data, keeping frequently used and safety-critical controls in a primary zone, and avoiding “bolt-on” panels that lead to clutter and mode confusion. These references became the backbone for decisions about steering height, screen bands, throttle location, alarm placement and the need to separate primary, secondary and tertiary zones on the console.

In parallel, case studies of Aditya (solar ferry), Kochi Water Metro vessels and MTCC/RTL high-speed ferries were reviewed through images, specs and available descriptions. Aditya and KWM provided a realistic picture of current Indian inland ferry helms: technically upgraded (electric/solar, integrated ticketing) but with largely conventional, sometimes cluttered wheelhouses and limited explicit human-factors framing. MTCC ferries added a contrasting high-speed, open-sea context with compact, response-focused helms. Across all three, the pattern

was similar: controls and screens had often evolved incrementally rather than from a single ergonomic concept. This reinforced the need for a more systematic, HFD-driven redesign in our project, where physical layout and digital dashboard are treated as one integrated cockpit rather than separate layers.

5.3.2 Primary Research

Primary research was conducted through informal interviews and contextual observation sessions with ferry captains, trainees and technical staff. Discussions covered everyday tasks (docking, departures, emergency handling), perceived pain points with the current helm, and expectations from a “better” console and dashboard. Whenever possible, the existing wheelhouse was observed during preparation or low-traffic operations to note actual hand movements, body postures, frequently used controls and workarounds. Notes and photographs from these sessions were used to validate secondary-research insights and to identify issues specific to the Kochi inland water context.

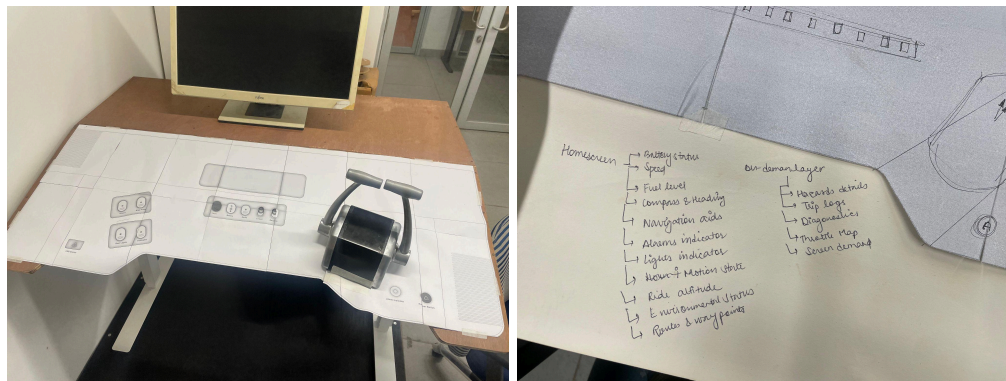


Fig 5.1 Initial research

5.3.3 HFD Data Collection - Anthropometry & Task Analysis

To ensure that the design fits a wide range of operators, relevant Indian anthropometric data was compiled, focusing on standing and seated eye height, vertical and horizontal reaches and leg clearance. These values were used to establish target bands for wheel height, display band, throttle position and console depth. In parallel, a task analysis was carried out for key scenarios such as “approach and docking”, “normal cruising” and “alarm response”. Each task was broken down into steps and mapped to the controls and information required, highlighting which actions are time-critical and which can tolerate more steps or greater reach.



Fig 5.2 Card Sorting & Ergonomics mapping

5.3.4 Research Synthesis Method

Findings from secondary and primary research were synthesised using simple affinity mapping and requirement clustering. Notes from interviews, observations and literature were written as individual statements and grouped into themes such as reach and posture, visibility, cognitive load, safety and digital complexity. From these clusters, explicit design requirements were derived—for example, “place high-frequency controls within primary reach zone” or “ensure all normal-operation information is available on a single main screen”. These requirements then fed directly into the generation of physical layout options and the information architecture for the digital dashboard.

5.4 Synthesis: Requirements & HFD Principles

The synthesis step translated research insights into a structured set of design requirements anchored in Human Factors in Design (HFD) principles. Requirements were organised under headings such as reach and posture, visual ergonomics, cognitive ergonomics and safety. For each heading, specific numeric targets or qualitative guidelines were defined and then used to evaluate and iterate physical and digital concepts.

5.4.1 Key findings from research

Across secondary and primary research, three key findings emerged. (i) First, existing helms tend to be cluttered and evolved through incremental additions, leading to scattered controls and mixed sight lines. (ii) Second, many critical controls and displays are not consistently located within comfortable reach or viewing angles for the full range of operators, especially shorter users. (iii) Third, digital systems are often layered on top of analogue consoles without a coherent information hierarchy, increasing cognitive load and the risk of missed or misunderstood alarms.

5.4.2 HFD Principles Applied

In response to these findings, several HFD principles were explicitly applied. Anthropometric fit and design for extremes guided all key heights and distances, targeting roughly the 5th to 95th percentile Indian operator. Reach zoning was used to organise controls into primary, secondary and tertiary areas based on frequency and criticality. Visual ergonomics principles informed the vertical placement and tilt of displays to keep critical information within a comfortable field of view and minimise glare. Cognitive ergonomics was addressed through grouping related functions, limiting mode changes and reducing the number of steps to access essential information or controls.

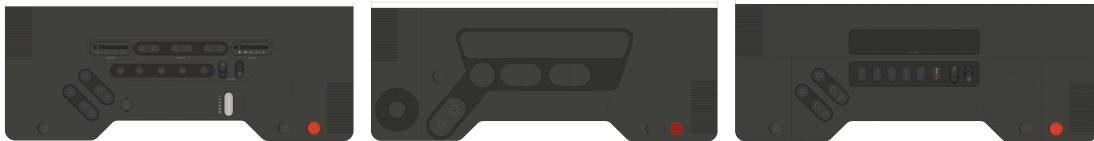


Fig 5.3 Initial Dashboard Outlook

5.4.3 Function Allocation: Physical vs Digital

A dedicated function-allocation step decided which actions should remain physical, which should live digitally and which should be mirrored in both. Steering, throttle and emergency actions such as the horn and emergency stop were kept as physical controls in the primary zone for reliability and fast, eyes-off activation. Monitoring and logging functions, detailed settings and non-critical alerts were shifted to the digital dashboard, where they could be layered and filtered more flexibly. Some functions, such as navigation status and key alarms, were intentionally mirrored: presented as physical indicators or buttons but also surfaced prominently on the main digital screens to support redundancy and situational awareness.



Fig 5.4 Functions & Details

5.5 Digital dashboard IA

The digital dashboard was structured using information architecture (IA) methods supported by a small card sorting exercise. All identified functions and data points were first listed and then tested with users to understand how they would naturally group and label them. The results informed a top-level navigation model and the distribution of content across the main screens. The aim was to ensure that routine, time-critical and safety-related information are surfaced with minimal steps while less frequently used tools remain accessible but unobtrusive.

A feature inventory was compiled by combining research findings, existing helm equipment and envisioned future capabilities. It included navigation-related functions (position, speed, route, depth, cameras), engine and energy monitoring (RPM, battery status, fuel, temperatures), communication (VHF, assistance requests), environmental data (weather, tides) and vessel services (lights, pumps, thrusters, HVAC).

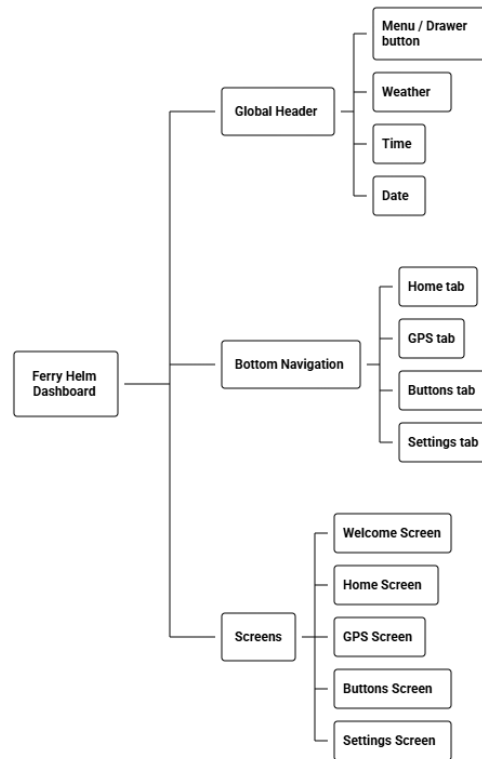


Fig 5.5 Information Architecture for the digital screens

At the top level, the dashboard is organised into a small set of tabs or modes—for example, Home, Navigation/GPS, Systems/Buttons and Settings—supplemented by a persistent header for global status information. Within each tab, functions are arranged in a shallow hierarchy so that important actions and readings are accessible in one or two steps. The final IA balances clarity and discoverability: it keeps routine and safety-critical operations prominent and consistent across screens, while still providing structured access to less frequent, detailed or configuration-oriented features

5.5.4 Visual Hierarchy & Screen design

In parallel, the project delivers a digital dashboard specification that aligns the information architecture with the physical control layout. Functions are clustered into navigation, engine monitoring, communication and auxiliary groups, and arranged into a shallow hierarchy so that routine and time-critical actions are reachable within one or two steps. Screen layouts specify the distribution of persistent status elements, contextual controls and alarm regions, with visual hierarchy tuned to support rapid scanning in both normal and off-normal conditions. The physical

and digital layers are explicitly mapped so that groups of on-screen functions correspond to coherent control clusters on the helm.

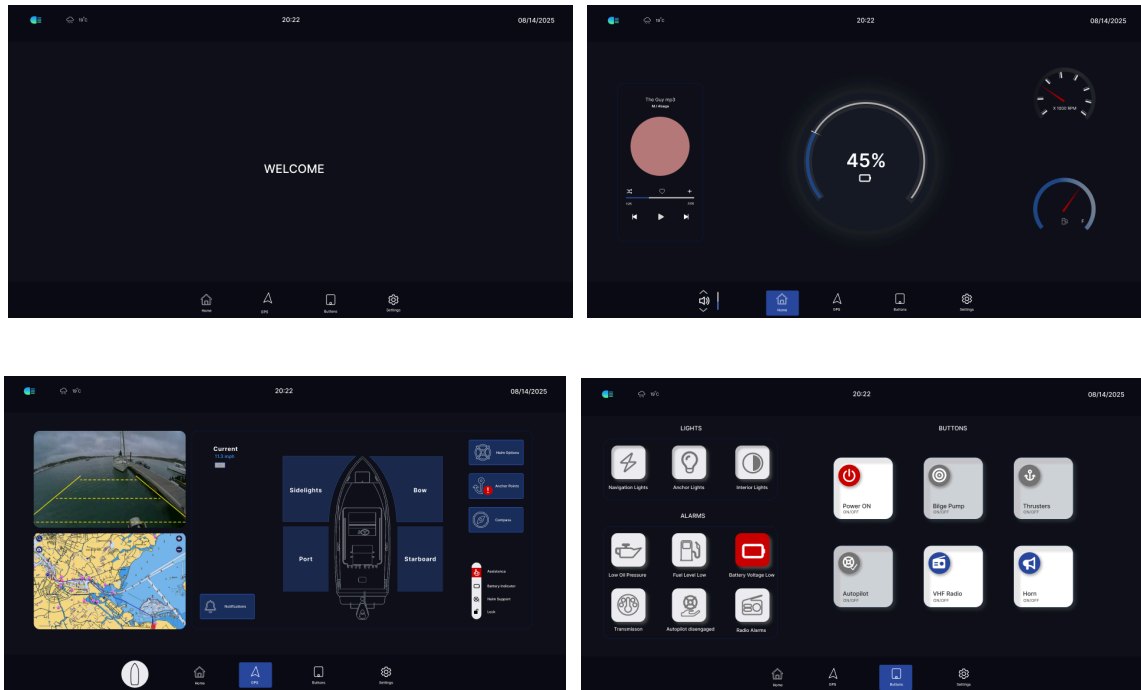


Fig 5.6 Key UI screens

5.6 Physical Cockpit

The first outcome is a fully resolved integrated helm architecture that organises steering, throttle and primary controls within clearly defined reach and visibility zones. Using Indian anthropometric data and task analysis, the console geometry is dimensioned so that high-frequency and safety-critical controls fall in the primary reach zone for both seated and standing operators, while low-frequency functions are shifted to secondary or tertiary zones.

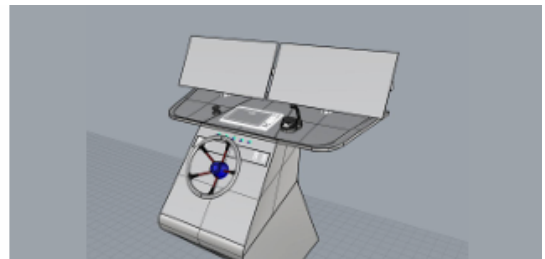
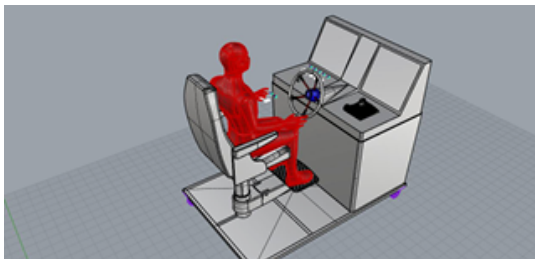




Fig 5.7 Views of initial design of the cockpit and final CAD

We began with exploratory CAD modelling, translating earlier hand sketches into multiple 3D variants of the console. At this stage, the focus was on overall volume, angles of the front face, and the relationship between the steering wheel, throttle, button clusters and screens. Several configurations were generated and compared against reach envelopes and sight-line requirements to eliminate forms that caused excessive stretching, glare or blocked views.

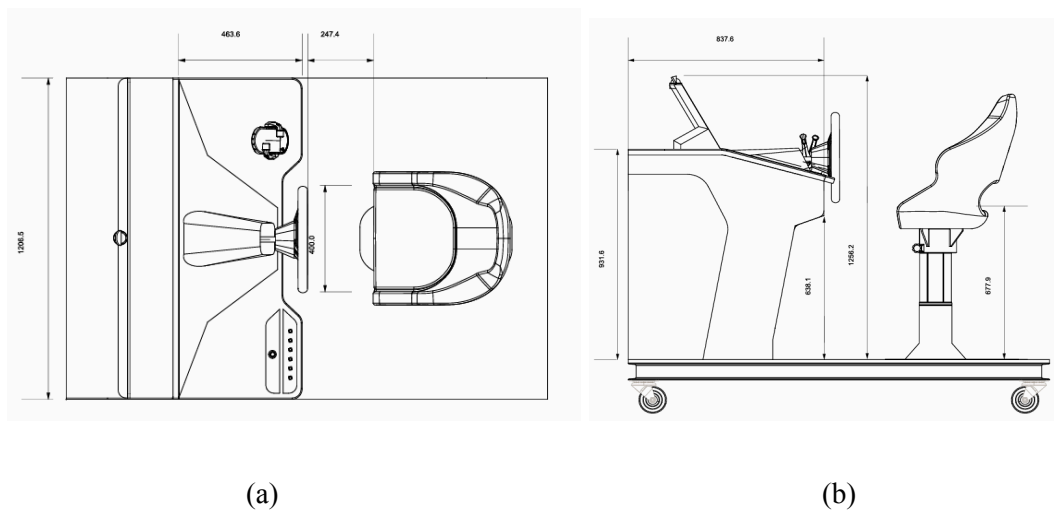


Fig 5.8 Dimensions of the helm (a) Top view (b) Side view

Based on this exploration, a final CAD configuration was selected and refined in detail. In this phase, surface transitions were cleaned up, panel breaks and control cut-outs were added, and the geometry was adjusted to hit the target dimensions defined in the ergonomics table (wheel height, screen band, throttle position, knee clearance).



Fig 5.9 Final CAD

The final CAD model specifies the overall envelope, mounting interfaces and grouped control zones, making it suitable as a basis for fabrication drawings and bill of materials.



Fig 5.10 Final mockup

The resolved CAD was then taken into the physical realm through a full-scale mock-up and subsequent prototype. The mock-up, built from low-cost materials, was used to quickly verify reach, posture and visibility with real users in both sitting and standing positions. Insights from this stage (for example, minor clashes, awkward corners or inconvenient button heights) were fed back into the CAD. A more robust prototype was then constructed to represent the final form and control layout.

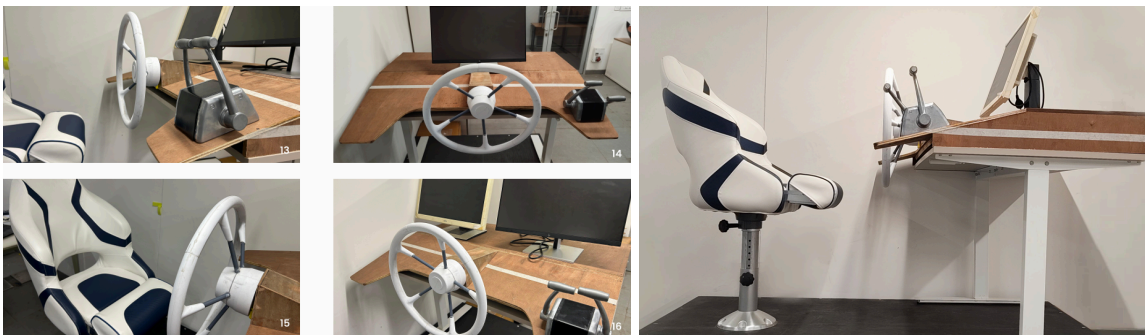


Fig 5.11 Placements on the prototype

ID	Dimension	Recommended (mm)	Design (mm)
H1	Wheel centre height	900 - 1,000	920
H2	Top of main display	1,050 - 1,200	990
H3	Throttle grip height	800 - 900	810
H4	Console width	900 - 1,300	1,200
H5	Console depth	500 - 800	300
H6	Primary reach radius (plan view)	350 - 450	420

Table 5.1 Key Helm Dimensions

5.7 Prototyping & Evaluation

A third major outcome is a set of physical and digital prototypes used to test the proposed cockpit in realistic conditions. Scale 3D-printed models were first used to assess overall form, proportion and the relative positioning of the wheel, throttle and screens within the wheelhouse envelope. These models helped check how far the console projects into the cabin, the angle of the front face and the visual relationship between the helm, windscreen and side windows before committing to a full-scale build.

On the basis of the final CAD, a full-scale mock-up was then constructed at the target dimensions defined in the ergonomics study (wheel height, screen band, throttle height, console depth and width as summarised in Table 5.1).



Fig 5.12 Final Prototype

During evaluation, participants of different heights sat and stood at the helm while key distances were observed: horizontal reach from shoulder to wheel and throttle, vertical distance from eye to top and bottom of the main display, knee and leg clearance under the console and side clearance for turning the wheel. Simple checks were used, such as whether primary controls could be reached without leaning forward, whether the display remained fully readable within a comfortable head movement, and whether there was sufficient space to move between sitting and standing without obstruction.

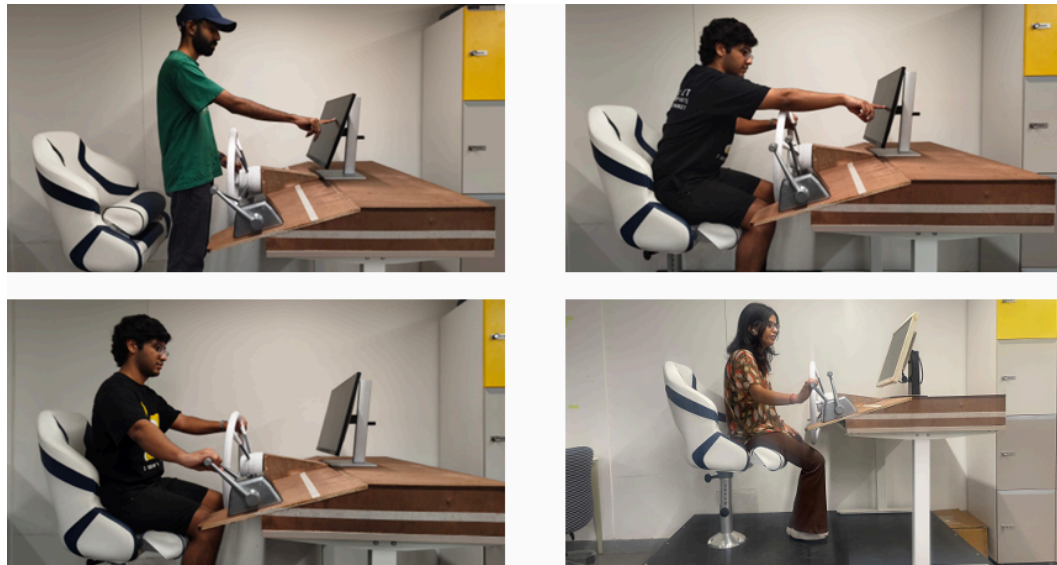


Fig 5.13 Ergonomics Test

In parallel, an interactive digital prototype of the dashboard was created to test information findability and menu logic. Participants were asked to complete short scenarios (e.g., “check fuel level”, “acknowledge an alarm”, “switch on navigation lights”, “view docking camera”), and for each task the number of taps, completion time and any hesitation points were noted. Combined with qualitative comments on visual clarity and perceived workload, these results fed back into small adjustments of card grouping, icon labels and the placement of high-priority functions on the Home, GPS and Buttons screens. Together, the physical and digital evaluations provided evidence that the proposed distances, reach zones and hierarchies are workable for a range of users before moving towards a more refined prototype.

Participant	Height Range	Posture	Main issue
Swattvik	5'4"	Sitting	Legs clear but steering wheel slightly far
Navatej	5'9"	Sitting and standing	Legs touch helm at closer setting
Anandhu	6'1"	Sitting and standing	Knees hit underside at close range, reach to the buttons were lesser

Table 5.2 – Ergonomic Test: Participant Summary

5.8 Discussion

The project demonstrates that applying Human Factors in Design systematically can significantly improve the usability of a ferry helm and dashboard compared to the existing, incrementally built consoles observed in current inland vessels. By grounding key dimensions in Indian anthropometric data and using reach zoning, the redesigned helm better accommodates a wider range of operators while reducing awkward postures and leg clashes identified in the full-scale mock-up tests. At the same time, the card-sorting–driven information architecture and restructured digital screens address the cognitive load created by scattered controls and unstructured displays, allowing critical navigation, engine and alarm information to be surfaced more clearly.

However, the work also has limitations. Evaluation was conducted with a limited number of participants, mainly peers and a small set of practitioners, and in a static lab environment rather than during live operations with vibration, glare and time pressure. The digital prototype focused on core flows and did not yet integrate live data from actual vessel systems. These constraints mean that while the results are promising and directionally robust, a further iteration cycle in collaboration with operators, manufacturers and regulators is needed before full adoption on working ferries.

6 Conclusion

This project set out to rethink the helm and digital dashboard of a Kochi Water Metro–type ferry from a human-factors perspective, treating the cockpit as an integrated physical–digital system rather than a collection of separate devices. Through a combination of standards-based secondary research, contextual primary research, anthropometric analysis, task breakdown and information architecture methods, it translates abstract HFD principles into a concrete console geometry, control layout and screen hierarchy. The resulting design improves reach, visibility and information clarity for a diverse range of Indian operators, while also proposing a cleaner, more future-ready framework for integrating additional functions over time.

Beyond the specific prototype, the main contribution lies in the process and guidelines: a repeatable way to allocate functions between physical and digital controls, to organise helm controls into primary/secondary/tertiary zones, and to structure marine dashboards around card-sorted function groups. These methods can be extended to other inland water transport systems, coastal ferries and even small workboats. Future work will involve in-situ trials on operational vessels, refinement of distances and surfaces based on long-term use, and exploration of more adaptive digital layouts that can respond to different crew profiles, conditions and modes of operation.

7 References

- [1]. ISO 8468:2007, *Ships and marine technology — Ship's bridge layout and associated equipment — Requirements and guidelines*. International Organization for Standardization.
- [2]. IMO MSC/Circ.982, *Guidelines on ergonomic criteria for bridge equipment and layout*. International Maritime Organization, 2000.
- [3]. American Bureau of Shipping, *Guidance Notes on the Ergonomic Design of Navigation Bridges*, 2018.
- [4]. Stopa, R. et al., "Bridge Ergonomic Design: A Review," *TransNav: International Journal on Marine Navigation and Safety of Sea Transportation*, 2017.
- [5]. Myles, H., *The Human Element in Ship Design*, The Nautical Institute, 2015.
- [6]. Rumawas, V., "A content analysis of human factors in ships' bridge design," *International Journal of Maritime Engineering*, 2014.
- [7]. Allsteel, *Ergonomics and Design: A Reference Guide*, 2004.
- [8]. INFLIBNET, "Anthropometry: Principles, measurements, application of anthropometry in ergonomics and design," Learning Module, 2020.
- [9]. NCSU Ergonomics Center, *Anthropometric Detailed Data Tables*, 2006.
- [10]. Taifa, I. W., and Desai, D. A., "Anthropometric measurements for ergonomic design of classroom furniture in India," *Journal of Engineering*, 2017.
- [11]. Karthik et al., "Anthropometric dimensions in standing and sitting postures for Indian agricultural workers," in tractor workplace design study, 2020.
- [12]. IJRESM, "Indian Anthropometric Dimension (Madhya Pradesh)," *International Journal of Research in Engineering, Science and Management*, 2018.
- [13]. Duran, I. et al., "Human–Machine Interaction Challenges for Bridge Operations," in *Proceedings of AHFE 2022*.
- [14]. Gaspar, J. F. et al., "Human centered design methodology: Case study of a ship bridge," *Ocean Engineering / Safety Science*, 2019.
- [15]. INTERTANKO, *Bridge Equipment and Layout Guidelines*, 2021.
- [16]. International Maritime Organization, *MSC/Circ.982 – Guidelines on Ergonomic Criteria for Bridge Equipment and Layout*, 2000.
- [17]. American Bureau of Shipping, *Guidance Notes on the Ergonomic Design of Navigation Bridges*, ABS, 2018 and updates

- [18]. Taifa, I. W., and Desai, D. A., “Anthropometric measurements for ergonomic design of engineering workplaces in India,” *Measurement*, 2017.
- [19]. “INDIAN Anthropometric Dimensions,” National Institute of Design / compiled report on Indian body dimensions, 1994.